

Gen AI Academy

APAC Edition

Build in APAC. Build for the world.



Participant Details

- a. Participant Name: Sejal Kshirsagar
- b. Problem Statement: Build a multi-agent AI system that helps users manage tasks, schedules, and information using multiple tools and data sources.

Brief about the idea

Smart Study Planner is a multi-agent AI system that helps students automatically create study plans, schedule tasks, and track progress.

It uses coordinated AI agents and modular tools to convert a simple input like “*I have an exam in 3 days*” into a structured, actionable workflow.

? Problem

Students struggle with:

- Planning study schedules
- Prioritizing topics
- Tracking progress
- Staying consistent

💡 Solution

We built a multi-agent AI system that:

- Generates structured study plans
- Assigns priorities (high/medium/low)
- Stores and schedules tasks
- Tracks completion and progress

⚙️ Approach (ADK + MCP)

- Used **ADK (Agent Development Kit)** to build agents
- Used **MCP-style tools** for:
 - Task storage
 - Calendar scheduling
 - Progress tracking
- Used **SQLite (lightweight DB)** for structured data

🌍 Real-World Impact

- Saves time for students
- Reduces planning stress
- Improves productivity
- Acts like a **personal AI study assistant**



Workflow

1. User gives input
2. Planner agent creates study plan
3. Task agent:
 - Stores tasks
 - Schedules tasks
4. User can:
 - View tasks
 - Mark tasks done
 - Track progress

Opportunities

★ What makes it different?

- Uses **multi-agent coordination**, not a single LLM
- Integrates **multiple tools (MCP-style)**
- Converts **natural language → real execution**

🔥 USP

- End-to-end workflow:
 - Planning → Scheduling → Tracking
- Modular architecture (easy to extend)
- Real-world usability (not just a chatbot)

✨ Features



AI-generated study plans



Priority-based scheduling



Calendar integration (mocked MCP tool)



Progress tracking



Task completion tracking

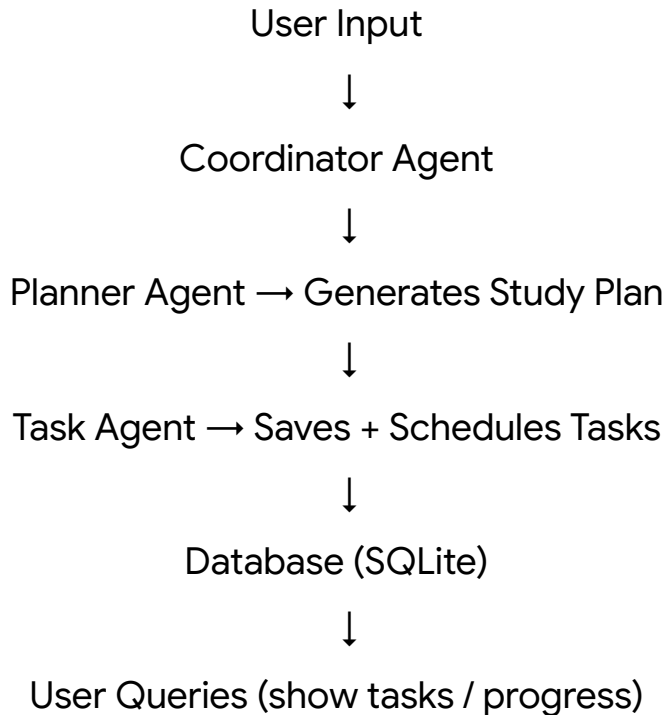


Natural language interaction



Multi-agent coordination

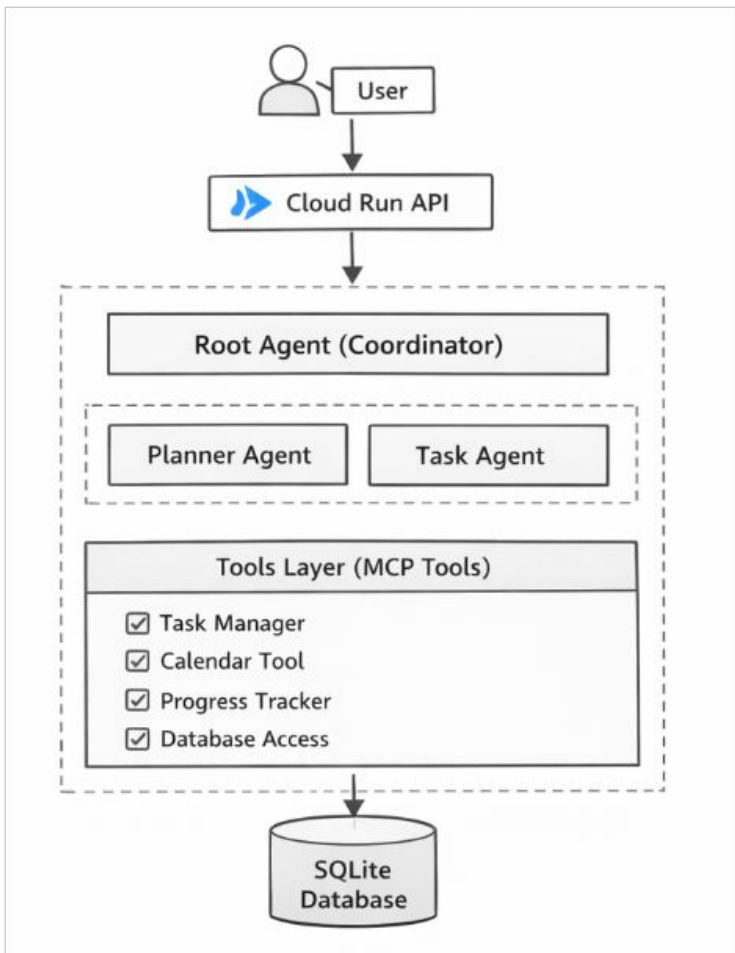
Process Flow Diagram



Wireframe



Architecture Diagram



Technologies/Google Services



Tech Stack

- Python
- Google ADK
- Gemini 2.5 Flash
- SQLite



Google Cloud

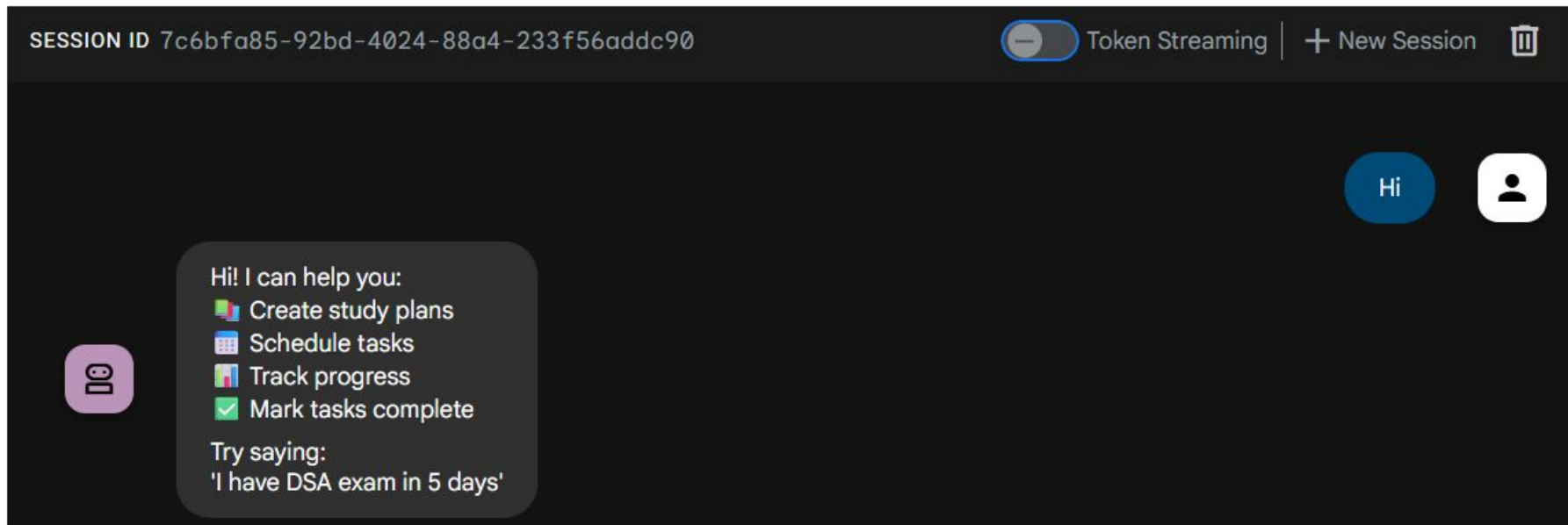
- Cloud Run → Deployment
- Vertex AI → LLM (Gemini)
- Artifact Registry → Container storage



Why this stack?

- **ADK** → easy multi-agent orchestration
- **Gemini Flash** → low cost, fast
- **Cloud Run** → scalable + serverless
- **SQLite** → lightweight for prototype

Snapshots of the prototype



SESSION ID 7c6bfa85-92bd-4024-88a4-233f56addc90



Token Streaming

+ New Session



I have a DSA exam in 3 days



⚡ transfer_to_agent



✓ transfer_to_agent



Day 1: Time & Space Complexity Analysis | high
Day 1: Arrays | low
Day 1: Linked Lists | medium
Day 1: Stacks & Queues | medium
Day 1: Recursion & Backtracking | high
Day 2: Trees (Binary Trees, BSTs) | high
Day 2: Hashing | medium
Day 2: Sorting Algorithms (Merge, Quick, Heap) | high
Day 2: Graphs (BFS, DFS) | high
Day 3: Graphs (Dijkstra, Kruskal, Prim) | high
Day 3: Dynamic Programming | high
Day 3: Trees (AVL/Red-Black, Heaps) | high



⚡ add_task



⚡ add to calendar

SESSION ID 7c6bfa85-92bd-4024-88a4-233f56addc90



Token Streaming

+ New Session



✓ add_task



✓ add_to_calendar



✓ add_task



✓ add_to_calendar



✓ add_task



✓ add_to_calendar



✓ add_task



✓ add_to_calendar



OK. I have added all those tasks to your study plan and calendar.

[show tasks](#)

DO

⚡ get_tasks

DO

✓ get_tasks

YOUR STUDY PLAN:

📅 Day 1

- Graphs (BFS, DFS, Dijkstra) | 🔥 high | pending
- Dynamic Programming (Basics, common patterns) | 🔥 high | pending
- Trees (Binary Trees, BSTs, Traversals) | 🔥 high | pending
- Recursion & Backtracking | 🔥 high | pending
- Time & Space Complexity Analysis | 🔥 high | pending
- Arrays | 🔥 low | pending
- Linked Lists | 🔥 medium | pending
- Stacks & Queues | 🔥 medium | pending
- Recursion & Backtracking | 🔥 high | pending

📅 Day 2

- Sorting Algorithms (Merge, Quick, Heap Sort) | 🔥 high | pending
- Hashing (Hash Maps, Collision Resolution) | 🔥 medium | pending
- Linked Lists (Operations, Two-pointer problems) | 🔥 medium | pending
- Arrays & Strings (Sliding Window, Two Pointers, Advanced) | 🔥 medium | pending
- Stacks & Queues (Application problems) | 🔥 medium | pending
- Heaps & Priority Queues | 🔥 medium | pending
- Trees (Binary Trees, BSTs) | 🔥 high | pending
- Hashing | 🔥 medium | pending
- Sorting Algorithms (Merge, Quick, Heap) | 🔥 high | pending
- Graphs (BFS, DFS) | 🔥 high | pending

📅 Day 3

- Graphs (Dijkstra, Kruskal, Prim) | 🔥 high | pending
- Dynamic Programming | 🔥 high | pending
- Trees (AVL/Red-Black, Heaps) | 🔥 high | pending

DO

The screenshot displays a dark-themed user interface for the Gen AI Academy. On the left, a vertical list of tasks is shown, each with a purple icon containing a white '10'. The tasks are:

- ⚡ mark_done
- ✓ mark_done
- ✓ Marked 'Arrays' as completed
- ⚡ get_progress
- ✓ get_progress

At the bottom left, a 'PROGRESS REPORT' box contains the following information:

- ✓ Completed: 1
- ✖ Total: 22
- 📊 Progress: 4%

On the right side of the interface, there are two buttons and user avatars:

- A blue button labeled 'mark Arrays done' with a user avatar.
- A blue button labeled 'progress' with a user avatar.



Gen AI Academy

APAC Edition

Thank you

Build in APAC. Build for the world.

